Homework 3. Logistic regression: The probability that a cab booking is cancelled

70-467 - Machine Learning for Business Analytics

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Background. YourCabs.com is a cab booking company in Bangalore, India. A customer books a cab by phone, on the website, or on the mobile site. The company then finds a car and a driver for that booking. Sometimes no car is available, or the driver backs out, and the company has to cancel the booking. When that happens close to the pickup time, the customer is left waiting for a car that never comes.

You have 10,000 bookings with pickup dates in 2013. Each observation is one booking. `Car_Cancellation` is 1 if the company cancelled the booking because no car was available, and 0 if it did not. About 7 in 100 bookings were cancelled. The dataset also records how the booking was made, the type of trip, the areas and map coordinates of the pickup and the drop-off, the car model, when the booking was made, and the pickup time. The codebook on the last page gives the type and the unit of every variable.

The data come from a contest YourCabs ran in 2013 with the Indian School of Business. This sample is the one used in the Taxi Cancellations case of Shmueli, Bruce, Gedeck, Yahav, and Patel, *Data Mining for Business Analytics* (Wiley).

The problem. Before a pickup, the operations desk can phone the driver to confirm, or line up a second car. It cannot do that for every booking. A booking the desk does not check, and that is then cancelled, leaves a customer stranded. A booking the desk checks that would have gone fine costs a phone call, and sometimes a car that waits for nothing.

The Head of Operations wants to know which bookings have a higher probability of being cancelled, and what about a booking is associated with that probability.

Please use the starter. Answer each question in the notebook, in the place where it asks.

The video. Imagine that you are recording a presentation of your analysis for the Head of Operations. The Head of Operations may consider your analysis when deciding which bookings the desk should check before pickup, what the desk should watch for when a new booking comes in, and how much higher the probability of cancellation is for those bookings. Use the answers you have already written.

Before you record, answer these for yourself. Then use what you decided in the presentation.

1. What decision will this person make with your probabilities?
2. What does the Head of Operations need to hear in order to use them, and what can stay in the notebook?
3. Which variables does the model use? Why did the others stay out?
4. How will you show the size of each difference in the probability, in percentage points, so the Head of Operations can tell whether to count on it? What will you set that difference next to?
5. When the marginal effect for one booking and the average marginal effect differ, which one are you using, and why?
6. How many bookings would you have the desk check, and what share of the cancellations would that reach?
7. Where could this person misuse your probabilities? The share cancelled was not the same in every month of 2013. What does that mean for using them in a later month?

8. Which graph carries that case? What does its title say, as a share cancelled or in percentage points? If you show another graph, can you say in a sentence or two how it bears on the choice?
9. How will you order the presentation so the decision is clear before the methods?

Deliverables.

Submit on Canvas by Sunday, 1 November 2026, 11:59 pm:

- a knitted R notebook (HTML or PDF from the qmd), with the code you add and your written answers,
- at most six slides, and
- a video of at most five minutes.

The notebook, the slides, and the video must use the same model and the same numbers.

The camera must be on. I need to see you delivering. Please refrain from reading the slides. That is a zero on the slides and the video. AI can assist you in preparing, but it should not be used to deliver the video. If you use AI to make the delivery for you, such as reading from a script, you will receive a zero for the video.

About AI. AI is allowed on this homework with caution, as described in the syllabus. Use it to understand the methods and to fix errors in your code, and say at the end of the notebook how you used it. The answers and the recommendation have to be yours, in your own words. If the work reads as if AI wrote the answers, I will do what I said at the start of the semester: I will ask you to come to my office and explain the whole homework to me. Depending on how that conversation goes, the grade on this homework can go as low as 0 and you could be reported for an academic integrity violation.

Please start early and give this your full effort. The homework is a great tool to help you apply and solidify the logistic regression we have covered so far in session 5. Please reach out if you need support.

Codebook. 10,000 observations, 19 variables. One observation = one booking. `Car_Cancellation` is the outcome.

Variable	Type	Unit	Description
row.	numeric	none	Observation number in this sample.
user_id	numeric	customer id	Customer who made the booking. 7,044 customers.
vehicle_model_id	numeric	model id	Car model requested. Model 12 is about 73% of bookings.
package_id	numeric	package id	Hourly rental package (hours and kilometers). Filled for hourly rentals only.
travel_type_id	numeric	1 / 2 / 3	1 = long distance, 2 = point to point within the city, 3 = hourly rental.
from_area_id	numeric	area id	Pickup area. Several hundred areas.
to_area_id	numeric	area id	Drop-off area. Point to point trips only.
from_city_id	numeric	city id	Pickup city. Missing on most bookings.
to_city_id	numeric	city id	Destination city. Long distance trips only.
from_date	text	m/d/yy h:mm	Requested pickup date and time.
to_date	text	m/d/yy h:mm	Requested end of the trip. Blank on many bookings.
online_booking	numeric	0 / 1	1 if the booking was made on the website.
mobile_site_booking	numeric	0 / 1	1 if the booking was made on the mobile site. If both are 0, the booking came by phone.
booking_created	text	m/d/yy h:mm	Date and time the booking was made.
from_lat, from_long	numeric	degrees	Map coordinates of the pickup area.
to_lat, to_long	numeric	degrees	Map coordinates of the drop-off area. Point to point trips only.
Car_Cancellation	numeric	0 / 1	1 if the company cancelled the booking because no car was available. This is the outcome.